

Nonhomogeneous linear systems

$$x'(t) = Ax(t) + f(t)$$

$$x(t) = x_H(t) + x_p(t)$$

2nd order
nonhomog.
linear DE
with const
coeff.

Undetermined coefficients

We find $x_p(t)$ in the form of $f(t)$

Ex: If $f(t) = e^{2t} \bar{a}$, then $x_p(t) = e^{2t} \bar{b}$ ← to find

If $f(t) = t^2 \bar{a}$, then $x_p(t) = \bar{b}t^2 + \bar{c}t + \bar{d}$

If $f(t) = (\sin t) \bar{a}$, then $x_p(t) = \bar{b} \sin t + \bar{c} \cos t$ are vectors

If $f(t) = \begin{bmatrix} t \\ e^t \\ t^2 \end{bmatrix}$, then $x_p(t) = \bar{a}e^t + \bar{b}t^2 + \bar{c}t + \bar{d}$

If $f(t) = \begin{bmatrix} t \\ \sin t \end{bmatrix}$, then $x_p(t) = \bar{a}t + \bar{b} + \bar{c} \sin t + \bar{d} \cos t$

Ex: $x'(t) = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} x(t) + t \begin{bmatrix} -9 \\ 0 \\ -18 \end{bmatrix}$

1) Solve homogeneous

$$|A - rI| = \begin{vmatrix} 1-r & -2 & 2 \\ -2 & 1-r & 2 \\ 2 & 2 & 1-r \end{vmatrix} = (1-r)^3 - 8 - 8 - 4(1-r) - 4(1-r) - 4(1-r) = 0$$

$$(1-r)^3 - 12(1-r) - 16 = 0$$

$$1-r = t \Rightarrow t^3 - 12t - 16 = 0 \Rightarrow (t+2)(t^2 - 2t - 8) = 0$$

$$\Rightarrow (t+2)(t-4)(t+2) = 0 \Rightarrow (t+2)^2(t-4) = 0 \Rightarrow (3-r)^2(-3-r) = 0$$

$$\Rightarrow r_1 = r_2 = 3, \quad r_3 = -3$$

$$(A - 3I)\bar{u} = 0 \Rightarrow \begin{bmatrix} -2 & -2 & -2 \\ -2 & -2 & -2 \\ -2 & -2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \Rightarrow x+y+z=0 \Rightarrow x = -y-z$$

$$y = s, \quad z = t$$

$$\text{then } \bar{u} = \begin{bmatrix} -s-t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \bar{u}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\bar{u}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$r_3 = -3$$

$$(A + 3I)\bar{u} = \begin{bmatrix} 4 & -2 & 2 \\ -2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \quad \begin{cases} 2x - y + z = 0 & (2) \times 2 + (1) \Rightarrow \\ -x + 2y + z = 0 & \end{cases}$$

$$\Rightarrow 3y + 3z = 0 \Rightarrow y = -z$$

$$(1) - (2) \Rightarrow 3x - 3y = 0 \Rightarrow x = y$$

$$y = s \Rightarrow \bar{u} = \begin{bmatrix} s \\ s \\ -s \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$x_H = c_1 e^{3t} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + c_3 e^{-3t} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

2) Find $x_p = \bar{a}t + \bar{b}$, where $\bar{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$, $\bar{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

$$x_p' = \bar{a} \Rightarrow \underline{\bar{a}} = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} (\underline{\bar{a}t + \bar{b}}) + t \underline{\begin{bmatrix} -9 \\ 0 \\ -18 \end{bmatrix}}$$

$$\bar{a} = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \bar{b}$$

and

$$\bar{0} = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \bar{a} + \begin{bmatrix} -9 \\ 0 \\ -18 \end{bmatrix}$$

$$\Rightarrow \begin{cases} a_1 = b_1 - 2b_2 + 2b_3 \\ a_2 = -2b_1 + b_2 + 2b_3 \\ a_3 = 2b_1 + 2b_2 + b_3 \end{cases}$$

and

$$\begin{cases} 0 = a_1 - 2a_2 + 2a_3 - 9 \\ 0 = -2a_1 + a_2 + 2a_3 \\ 0 = 2a_1 + 2a_2 + a_3 - 18 \end{cases}$$

$$\begin{cases} a_1 - 2a_2 + 2a_3 = 9 \\ -2a_1 + a_2 + 2a_3 = 0 \\ 2a_1 + 2a_2 + a_3 = 18 \end{cases}$$

$\Downarrow$

$$\begin{cases} a_1 - 2a_2 + 2a_3 = 9 \\ -3a_2 + 6a_3 = 18 \\ 6a_2 - 3a_3 = 0 \end{cases}$$

$\Downarrow$

$$\begin{aligned} a_1 - 2a_2 + 2a_3 &= 9 \\ -a_2 + 2a_3 &= 6 \\ 9a_3 &= 36 \end{aligned}$$

$$\begin{cases} a_1 = 5 \\ a_2 = 2 \\ a_3 = 4 \end{cases}$$

$$\text{and } \begin{cases} b_1 - 2b_2 + 2b_3 = 5 \\ -2b_1 + b_2 + 2b_3 = 2 \\ 2b_1 + 2b_2 + b_3 = 4 \end{cases}$$

$\Downarrow$

$$\begin{cases} b_1 - 2b_2 + 2b_3 = 5 \\ -3b_2 + 6b_3 = 12 \\ 6b_2 - 3b_3 = -6 \end{cases}$$

$\Downarrow$

$$\begin{aligned} b_1 - 2b_2 + 2b_3 &= 5 \\ -b_2 + 2b_3 &= 4 \\ 9b_3 &= 18 \end{aligned}$$

$$\begin{cases} b_1 = 1 \\ b_2 = 0 \\ b_3 = 2 \end{cases}$$

$$\Rightarrow X_p = \begin{bmatrix} 5 \\ 2 \\ 4 \end{bmatrix} t + \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$\Rightarrow X = X_h + X_p$$

Variation of Parameters

$$X'(t) = A X(t) + f(t) \quad (1)$$

1) Solve $X'(t) = A X(t)$, get $X(t) \leftarrow$ fundamental matrix

$$\Rightarrow X_h(t) = X(t) \bar{c}, \quad \bar{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$$2) X_p(t) = X(t) \bar{u}(t)$$

$$\text{Sub into (1)} \Rightarrow \cancel{X'(t) \bar{u}(t)} + X(t) \bar{u}'(t) = A \cancel{X(t) \bar{u}(t)} + f(t)$$

$$\text{We have that } X'(t) = A X(t)$$

$$\Rightarrow X(t) \bar{u}'(t) = f(t) \Rightarrow \text{find } X^{-1}(t) \swarrow \begin{array}{l} \text{exists} \\ \text{since} \\ \text{the col.} \\ \text{are lin} \\ \text{indep.} \end{array}$$

$$\underbrace{X^{-1}(t) X(t)}_I \bar{v}'(t) = X^{-1}(t) f(t)$$

$$\Rightarrow \bar{v}'(t) = X^{-1}(t) f(t) \Rightarrow \bar{v}(t) = \int X^{-1}(t) f(t) dt$$

$$\Rightarrow \boxed{x_p(t) = X(t) \int \underline{X^{-1}(t)} f(t) dt}$$

Ex: $x'(t) = \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix} x(t) + \begin{bmatrix} e^{2t} \\ 1 \end{bmatrix}$

1) $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & -3 \\ 1 & -2-\lambda \end{vmatrix} = -4 + \lambda^2 + 3 = \lambda^2 - 1 = 0$
 $\Rightarrow \lambda_1 = 1, \lambda_2 = -1$

$(A - I)\bar{u}_1 = \begin{bmatrix} 1 & -3 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = 0 \Rightarrow x_1 - 3y_1 = 0 \Rightarrow \bar{u}_1 = \begin{bmatrix} 3s \\ s \end{bmatrix} = s \begin{bmatrix} 3 \\ 1 \end{bmatrix}$
 $x_1 = 3y_1, y_1 = s$

$(A + I)\bar{u}_2 = \begin{bmatrix} 3 & -3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = 0 \Rightarrow x_2 = y_2 \Rightarrow \bar{u}_2 = s \begin{bmatrix} 1 \\ 1 \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\Rightarrow x_1(t) = e^t \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad x_2(t) = e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow x(t) = \begin{bmatrix} 3e^t & e^{-t} \\ e^t & e^{-t} \end{bmatrix}$

$$\left[\begin{array}{cc|cc} 3e^t & e^{-t} & 1 & 0 \\ e^t & e^{-t} & 0 & 1 \end{array} \right] = \left[\begin{array}{cc|cc} 3e^t & \textcircled{e^{-t}} & 1 & 0 \\ 0 & -2e^{-t} & 1 & -3 \end{array} \right] =$$

$$= \left[\begin{array}{cc|cc} 6e^t & 0 & 3 & -3 \\ 0 & -2e^{-t} & 1 & -3 \end{array} \right] = \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{2}e^{-t} & -\frac{1}{2}e^{-t} \\ 0 & 1 & -\frac{1}{2}e^t & \frac{3}{2}e^t \end{array} \right]$$

$$\Rightarrow X_p = \left[\begin{array}{cc} 3e^t & e^{-t} \\ e^t & e^{-t} \end{array} \right] \int \left[\begin{array}{cc} \frac{1}{2}e^{-t} & -\frac{1}{2}e^{-t} \\ -\frac{1}{2}e^t & \frac{3}{2}e^t \end{array} \right] \left[\begin{array}{c} e^{2t} \\ 1 \end{array} \right] dt = \underbrace{\hspace{10em}}_{X^{-1}(t)}$$

$$= \left[\begin{array}{c} \downarrow \\ \int \left[\begin{array}{cc} \frac{1}{2}e^t - \frac{1}{2}e^{-t} \\ -\frac{1}{2}e^{3t} + \frac{3}{2}e^t \end{array} \right] dt = \left[\begin{array}{c} \frac{1}{2}e^t + \frac{1}{2}e^{-t} \\ -\frac{1}{6}e^{3t} + \frac{3}{2}e^t \end{array} \right] \end{array} \right]$$

$$= \begin{bmatrix} 3e^t & e^{-t} \\ e^t & e^{-t} \end{bmatrix} \begin{bmatrix} \frac{1}{2}e^t + \frac{1}{2}e^{-t} \\ -\frac{1}{6}e^{3t} + \frac{3}{2}e^t \end{bmatrix} = \begin{bmatrix} \frac{3}{2}e^{2t} + \frac{3}{2} - \frac{1}{6}e^{2t} + \frac{3}{2} \\ \frac{1}{2}e^{2t} + \frac{1}{2} - \frac{1}{6}e^{2t} + \frac{3}{2} \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{4}{3}e^{2t} + 3 \\ \frac{1}{3}e^{2t} + 2 \end{bmatrix} = x_p$$

$$X = X_H + X_p = c_1 e^t \begin{bmatrix} 3 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{3} e^{2t} \begin{bmatrix} 4 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$